Project Initialization and Planning Phase

Date	28 October 2026
Team ID	
Project Title	Health360: Comprehensive Healthcare Risk & Disease Analytics
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	To build an end-to-end healthcare analytics ecosystem that processes 5,000 patient records across clinical vitals (blood pressure, glucose, BMI) and lifestyle factors to deliver early risk stratification, interactive Tableau dashboards, and a web-embedded executive interface.
Scope	The project encompasses dataset cleaning and feature engineering in Python (calculating Age Brackets, WHO BMI Categories, and Dynamic Risk Tier Index), designing 9 visual worksheets and 2 master dashboards in Tableau, constructing a 4-scene Executive Story, and embedding the analytics into a Python Flask web application (app.py) via Tableau JS API v2.
Problem Statement	
Description	Healthcare providers struggle with fragmented patient data silos, making it difficult to detect early-stage chronic disease risks or analyze how lifestyle choices (smoking, alcohol, lack of exercise) compound clinical vitals like hypertension and elevated glucose.
Impact	Unidentified high-risk patients suffer from delayed clinical interventions, leading to increased hospitalization rates, higher treatment costs, and suboptimal hospital resource allocation.
Proposed Solution	

Approach	Combine Python data engineering, Tableau business intelligence visualization, and full-stack web integration. We engineer dynamic risk indices from 5,000 clinical records, visualize cross-variable risk patterns in Tableau, and serve the interactive analytics locally over a lightweight Flask web framework (http://127.0.0.1:5001).
Key Features	1. Dynamic Risk Tier Indexing (High, Moderate, Low Risk). 2. Cohort Demographics & Age Stratification (18–34 to 65+). 3. Multi-Variable Clinical Matrix & Symptom Burden Heatmap. 4. Interactive 4-Scene Tableau Executive Story Container. 5. Web Integration via Flask & Tableau JavaScript API v2.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Quad-Core Processor (standard workstation / PC)
Memory	RAM specifications	Minimum 8 GB RAM
Storage	Disk space for data, models, and logs	500 MB free storage space
Software		
Frameworks	Python frameworks	Python 3.14, Flask, Tableau Desktop / Tableau Public
Libraries	Additional libraries	Pandas; HTML5, CSS3, JavaScript; Tableau JavaScript API v2
Development Environment	IDE, version control	VS Code; Google Chrome / MS Edge
Data		
Data	Source, size, format	smart_healthcare_dataset.csv, 5,000 anonymized patient records, 16 fields, CSV format
Personnel		

Roles	Team members and responsibilities	1 Data Engineer & BI Developer (Harshdeep Singh Sisodiya): end-to-end feature engineering, dashboard design, web hosting, and project documentation

Figure 1: Health360 Proposed Solution Architecture & Web Interface

